Krishnaa **Vadivel** | Semiconductor Physics | Integrated Photonics | Thin-Film Fabrication

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Profile

Electrical engineering PhD researcher with broad preparation across thin-film fabrication, semiconductor physics, integrated photonics, device simulation, and scientific software. Combines cleanroom and characterization experience with light-matter modeling, distributed HPC software, and machine learning for semiconductor and photonic systems.

Research Experience

Yale University, Electrical Engineering

Research Assistant / PhD Researcher *2021–present*

- Developed GaN-based power-electronics devices in cleanroom and MOCVD environments during the early PhD, including fabrication-oriented process work, selective-area etching techniques, and materials development.
- Characterized semiconductor materials with SEM, cathodoluminescence, and Raman spectroscopy, connecting fabrication outcomes to optical and structural behavior.
- Built simulation tools for VCSEL emission and coupled photon, phonon, and electron interactions in semiconductor and photonics problems.
- Developed distributed HPC software for exciton-polaron and light-matter interaction modeling using first-principles methods, quantum many-body theory, MPI, OpenMP, CUDA, ROCm, PETSc, and SLEPc.
- Used agentic ML and retrieval-based tooling to generate first-principles software inputs from journal literature and technical documentation.
- Maintained strong continuity between theory and engineering practice through coursework and self-developed notes in electromagnetism, signals/systems, semiconductor transport, optics, and device equations.
- Co-authored a 2025 *Journal of the American Chemical Society* paper and presented related APS work in 2024, 2025, and 2026 on self-trapped excitons and excited-state materials phenomena.

Professional Experience

Probe Technologies Inc. / Weatherford Wireline Services

Software and Embedded Systems Engineer *2019–2021*

- Parallelized CUDA-based software for analysis and visualization of acoustic and nuclear well-tool data used in engineering diagnostics and field interpretation.
- Applied dolfinx-based finite-element analysis to frequency studies of steel pipes in well environments to support physics-driven engineering analysis.
- Contributed instrumentation-oriented engineering around deployed well tools, connecting field data, sensing systems, and numerical analysis.

FindLight

Photonics Technical Writer *2018–2019*

- Produced technical articles on lasers, photonics components, optical systems, and engineering applications for a specialized industrial audience.

Selected Projects

Integrated Photonics Band-Pass Filter

- Designed a Meep-based integrated photonics band-pass filter, then fabricated and tested the device with an erbium-doped fiber laser setup.

Semiconductor Device Simulation

- Developed numerical treatments of Poisson, drift-diffusion, and pn-junction-style device equations to connect semiconductor theory with engineering-oriented simulation.

VCSEL Light-Matter Modeling

- Built simulation tools for VCSEL emission and coupled photon, phonon, and electron interactions in semiconductor systems.

Equivariant Neural Networks for Materials and Optics

- Developed PyTorch Geometric equivariant neural networks for material energy prediction and optical-property prediction.

Education

Yale University
PhD, Electrical Engineering *2021–present*
Ongoing research spanning computational materials, semiconductor-relevant modeling, and scientific software.

Yale University
MPhil, Electrical Engineering *2023*

Yale University
MS, Electrical Engineering *2023*

Cornell University
MEng, Electrical and Computer Engineering *2017–2018*
Included work in stochastic computing, semiconductor/device topics, and advanced EE design.

Cornell University
BS, Electrical and Computer Engineering *2013–2017*
Included strong preparation in embedded systems, robotics, controls, and hardware-software integration.

Selected Coursework

Semiconductors: Semiconductor Physics; Thin Film Science; drift-diffusion and device modeling foundations

Photonics: Lasers and Optoelectronics; Integrated Photonics

Circuits and VLSI: VLSI Design; Analog VLSI Design; stochastic-computing-related design work

Systems: Mathematics of Signals and Systems; Advanced Embedded Systems

Foundations: Graduate Quantum Mechanics; Solid State Physics; Many-Body Quantum Physics

Technical Skills

Languages: Python, C, C++, CUDA, JavaScript, Bash, PowerShell

Fabrication / Characterization: Cleanroom processing, MOCVD, SEM, cathodoluminescence, Raman spectroscopy, laser-bench optical testing

Simulation: Semiconductor device modeling, FDTD photonics simulation, finite-element analysis, first-principles modeling, quantum many-body modeling

Machine Learning: PyTorch, PyTorch Geometric, equivariant graph neural networks, materials- and optical-property prediction, agentic input generation

Scientific Computing: MPI, OpenMP, CUDA, ROCm, PETSc, SLEPc, NumPy, pandas, SciPy, SymPy, matplotlib

Software / Platforms: Python, C, C++, C#, ASP.NET, Microsoft SQL Server, Linux command line, Docker, Docker Compose, Kubernetes, gcloud, Frontier, Frontera, Perlmutter